

Healthcare Resource Availability: Analyzing Global Disparities

ADAN8880 - Directed Practicum in Analytics

Professor: Nurtekin Savas

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Group Members: Jen Jiang, Yiwen Ma

MDLC Report

Problem definition

This project aimed to predict life expectancy across 33 countries using publicly available World Bank data. The goal was to analyze global healthcare disparities and identify patterns associated with socio-economic status, public health infrastructure, and healthcare investments. We sought to support evidence-based discussions around global health equity.

Data acquisition

We used longitudinal data from the World Bank covering the years 2002 to 2021. The dataset included over 50 indicators related to healthcare resources, expenditures, outcomes, public health, WASH (water, sanitation, hygiene), and socio-demographics. Countries were grouped by income level (low, middle, high). Our target variable was life expectancy at birth (total, years). We split the dataset into training (2002–2015), validation (2016–2018), and test sets (2019–2021).

Data preparation

We cleaned the data by handling missing values and removing non-numeric columns. To improve model performance and generalization, we engineered several new features:

- Country level averages to capture long-term trends
- Income group year averages to capture macroeconomic context
- Temporal delta features to reflect change over time

We also identified proxy bias features that were highly correlated with income group and excluded them in the bias-mitigated model.

Model development

We experimented with multiple regression models, including Random Forest, LightGBM, and XGBoost, each tested under three configurations: a baseline version, a feature-engineered version, and a bias-mitigated version.

After evaluating all combinations, we found that XGBoost with bias mitigation provided the best overall balance of predictive performance and fairness.

We then focused on two final XGBoost models:

- Original model: included all numeric features
- Bias-mitigated model: excluded proxy features that could introduce income-related bias

Both models used the same hyperparameters to ensure a fair comparison, and were trained using the cleaned and feature-engineered dataset.

Model evaluation

All models were evaluated using RMSE and R^2 on the validation set. We also assessed fairness across income groups to detect any bias in prediction performance. Among all tested models and variations, the bias-mitigated XGBoost model achieved the best overall results:

- RMSE: 0.0661 (lowest among all models)
- R^2 : 0.9999
- Fairness: more balanced performance across high, middle, and low-income countries compared to non-mitigated models

We also used SHAP values to interpret feature contributions at both global and individual levels. This helped validate the model's reasoning and ensured transparency, particularly around sensitive socioeconomic indicators.

Model deployment

The selected model, bias-mitigated XGBoost, was saved as a pickle file (`xgb_mitigated_model.pkl`) for production use. Batch deployment was chosen, as life expectancy data is updated annually and does not require real-time inference.

Deployment planning included:

- Saving the trained model using pickle
- Documenting the environment
- Structuring the GitHub repo with Reports/, Notebooks/, and Models/ folders

Model monitoring & maintenance

We developed a monitoring strategy based on RMSE thresholds and prediction stability over time:

- Green: RMSE < 2.0: no action required
- Yellow: RMSE 2.0–3.0: review the model and monitor closely
- Red: RMSE > 3.0: pull the model from production and retrain

We also planned for data drift and concept drift, especially given the potential impact of global crises. The model will be retrained annually or sooner if drift is detected. Fairness across income groups will continue to be evaluated during retraining cycles.